

OMM Lecture: Basics of Dx: Innominates and pubic bone

PPOM1

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Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

1. Describe the basic anatomy of the pelvis and explain how it relates to the biomechanics of this region.
2. Describe and discuss the standing flexion test, ASIS compression test, and the seated flexion test, including how to perform them, what a positive finding is in each, and what this means osteopathically for diagnosing innominate and pubic somatic dysfunction.
3. Compare and contrast the standing flexion test and seated flexion tests.
4. Identify the landmarks that are used to perform to perform the innominate and pubic bone screening tests

You will learn about diagnosing the various specific pelvic (ie. innominate and pubic) somatic dysfunctions in the pelvis lab.

Outline

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I. Structure:

- a. Bones, Joints, Ligaments, Muscles
- b. Nervous System, Viscera

II. Function:

- a. Respiration and Circulation
- b. Primary Respiration
- c. Gait
- d. Obstetrical consideration

III. Somatic dysfunction

- a. Effect on Function

IV. Structural Exam

- a. Motion Testing: Standing Flexion Test, ASIS Compression Test (for pelvis)
- b. Seated flexion test (for sacrum)
- c. Static Exam

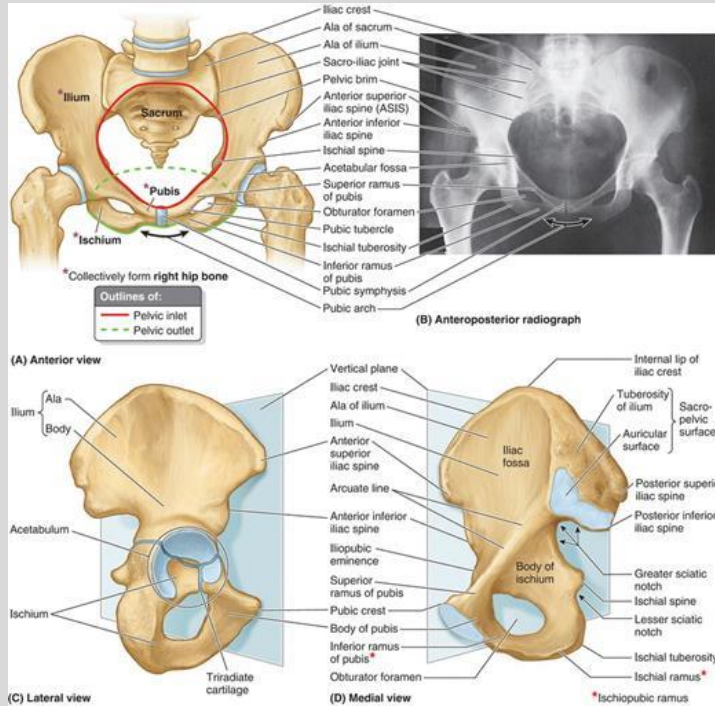
V. Practice Questions

- The following slides (5-18) on anatomy are to give you an overview of the functional significance of the pelvis. Aim to understand the general points, but you do not have to worry about memorizing details of the anatomy for the purposes of this lecture.

Structure: Bones

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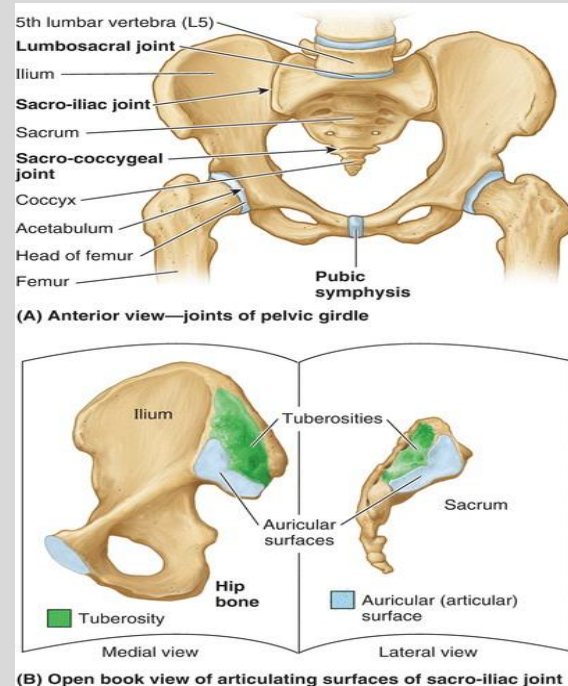
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- 3 parts at birth
 - Ilium
 - Ischium
 - Pubes
- Parts come together at acetabulum

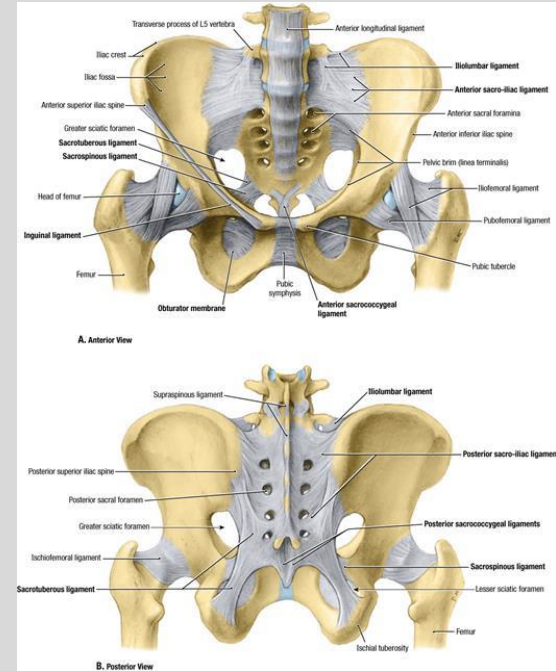
Structure: Joints

- Joints allow for mobility
- SIJ (sacroiliac joint)
 - Males: ligaments are well developed
 - Females: allow for mobility
 - Motion is more and more limited as one ages
 - L- or C-shaped
- Pubic Symphysis
 - Fibrocartilagenous joint



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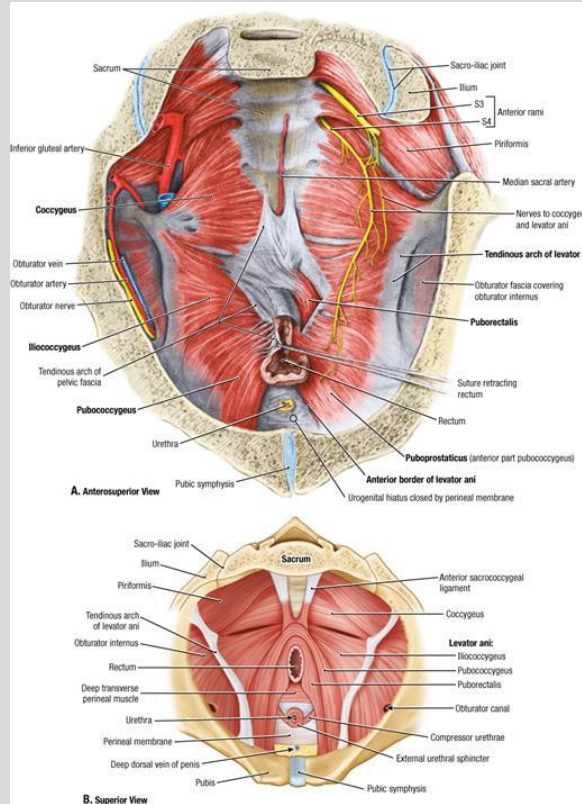
- The pelvis connects the lower extremities to the spine and provides a foundation for body support and locomotion
- Motion occurs around the sacroiliac joints and pubic symphysis. Examples are:
 - ❖ During the gait cycle
 - ❖ During flexion and extension of the spine; Sidebending and rotation also impact the SI joints
 - ❖ During childbirth



Agur et al. Grant's Atlas of Anatomy, 15e, Chp. 5, Pelvis and Perineum,
<https://login.nyit.idm.oclc.org/login?url=https://premium.basicsciences.lwwhealthlibrary.com/content.aspx?sectionid=239847266&bookid=2827#239847292>

Structure: Muscles

- Muscles of the pelvis provide support for the contents
- Pelvic Floor
 - Pelvic Diaphragm
 - Levator ani
 - Coccygeus
- Pelvic Wall
 - Piriformis
 - Obturator internus

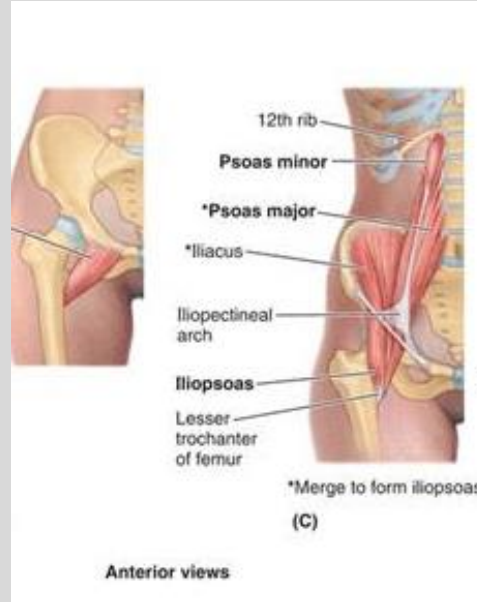


Agur et al. Grant's Atlas of Anatomy, 15e, Chapter 5

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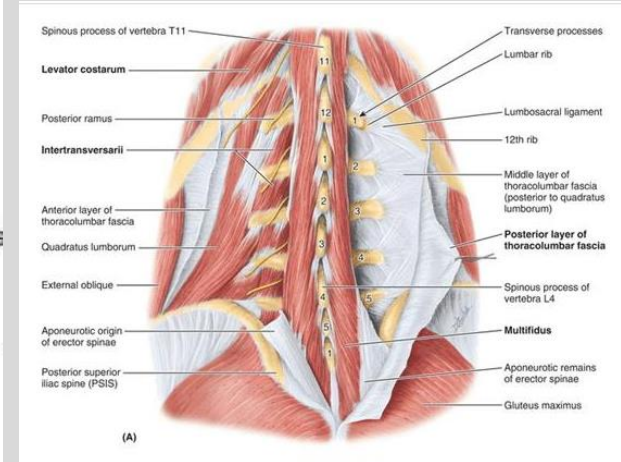
Structure: Muscles

- Muscles of the trunk and lower extremity allow for locomotion
- Rectus abdominis
- Transverse abdominis
- Internal and external oblique
- Quadratus lumborum
 - pain in low back pain, pelvis, etc.
- Iliopsoas
 - pain in low back pain, hip, pelvis, groin, etc.



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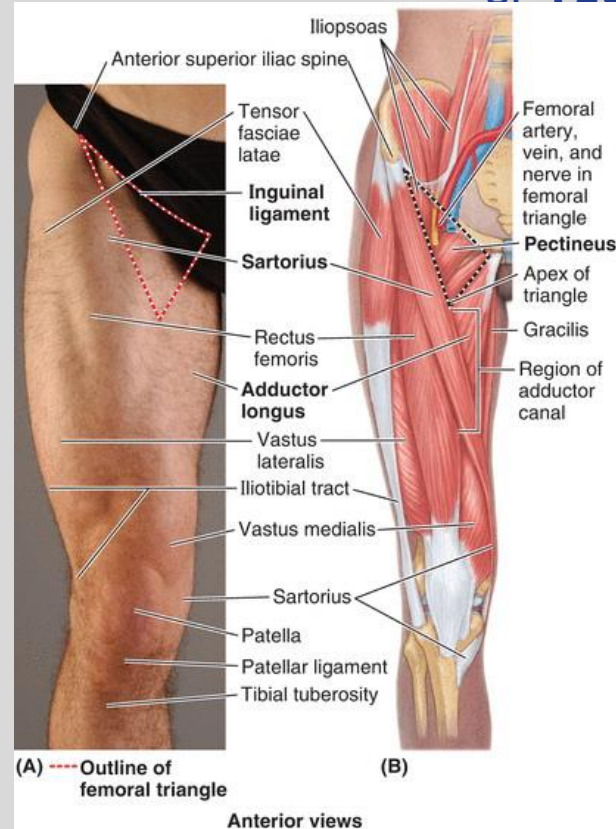
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Moore et al, Clinically Oriented Anatomy, 8e, Chp. 2
(Back) and Chp. 7 (Lower limb)

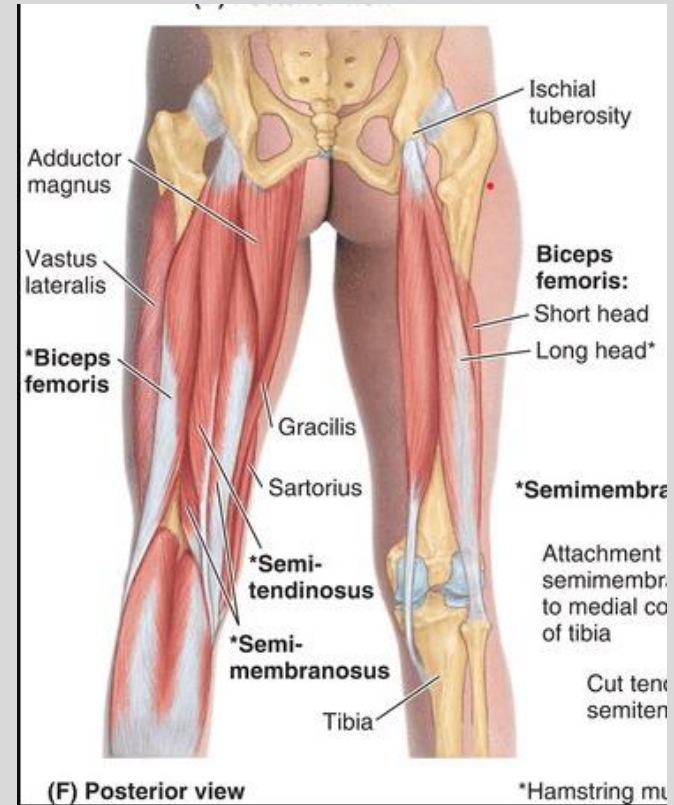
Structure: Muscles of Lower Extremity (Anterior)

- Rectus Femoris
 - Origin: AIIS
- Sartorius
 - Origin: ASIS
- Gracilis
- TFL
- Iliopsoas
- Adductors
 - Origin: Pubic bone



Structure: Muscles of Lower Extremity (Posterior)

- Gluteus max., med., min.
- Adductors
- Hamstrings (biceps femoris, semimembranosus, semitendinosus)
 - Origin: Ischial tuberosity



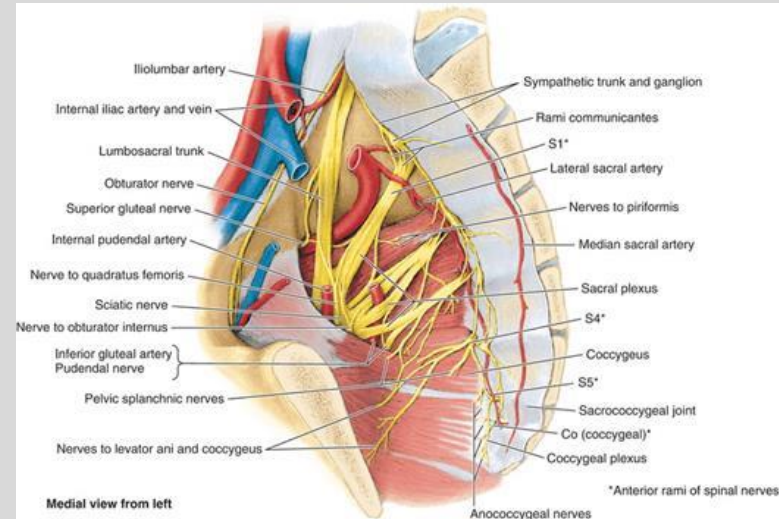
Structure: Nervous System

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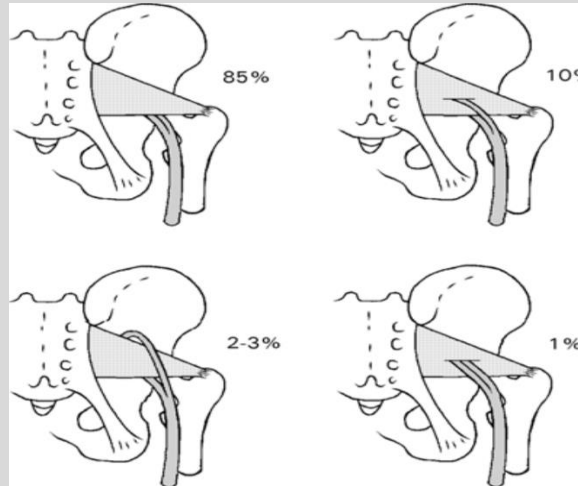
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Physiological

- Sacral sympathetic trunks
 - Extensions of lumbar sympathetic chain ganglia
 - Fuse over the coccyx to form the single **ganglion impar**
- Parasympathetic nerves of the pelvic splanchnic
 - Arise directly off ventral rami of **S2-4** and supply parasympathetic innervation to the *left colon* and *inferior hypogastric plexus*
- Inferior hypogastric plexus
 - Both parasympathetic and sympathetic fibers
 - Smaller plexi to the rectum, bladder, prostate, uterus, and vagina



Moore et al, Clinically Oriented Anatomy 8e, Chp. 6
Pelvis and Perineum



Legend:

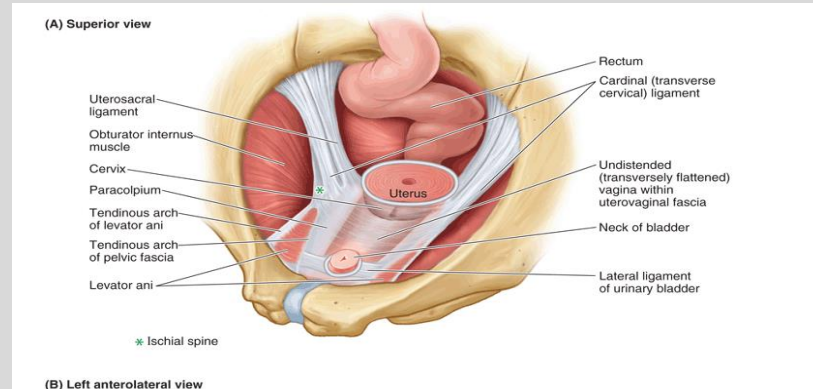
Position of sciatic nerve in relationship to the piriformis muscle. (Adapted from Beaton LE, Anson BJ. The sciatic nerve and the piriformis muscle: their interrelationship as a possible cause of coccygodynia. J Bone Joint Surg (Br) 1938;20:686-688.)

Structure: Viscera

- The pelvis is a container for pelvic organs
 - GI: rectum
 - GU: bladder
 - Repro: uterus, ovaries, vagina; prostate, penis

Structure: Viscera

- Ligaments (fascial condensations)
 - Carry vessels and nerves
 - Support and connect pelvic organs to the walls of the pelvis
- Lateral ligament of the bladder
- Cardinal ligament (base of Broad ligament)
- Uterosacral ligaments



Moore et al, Clinically Oriented Anatomy, 8e, Chp. 6 Pelvis and Perineum

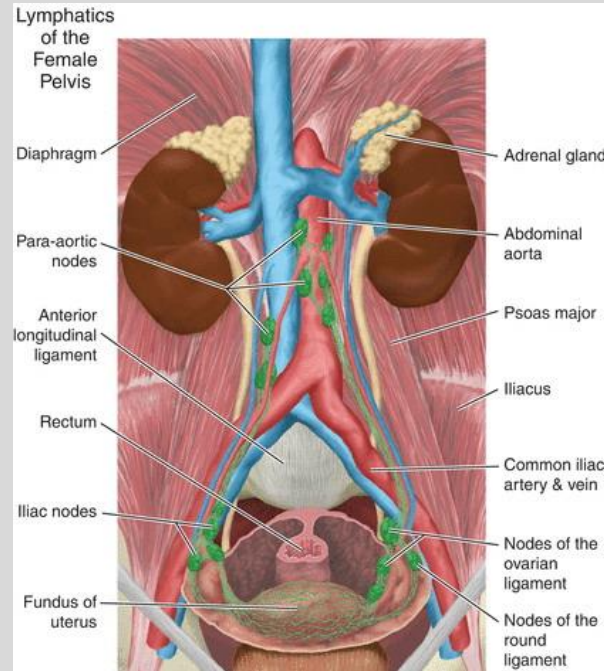
Function: Respiration, Circulation

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- Respiration: Abdominal diaphragm & pelvic diaphragm work together for breathing
- Circulation: Arteries, veins, nerves, lymphatic supply

➤ **Pelvic motion, free from restriction, is important for optimal function**



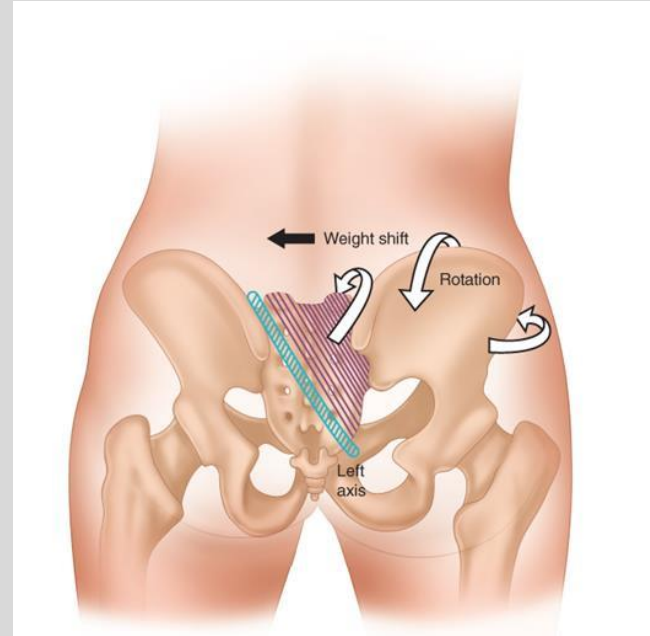
Seffinger et al., Foundations of Osteopathic Medicine 8e, Chp. 8,
Anatomy and Physiology of the Lymphatic System

Function: Gait Cycle

- During normal walking, physiologic motions of the sacrum and ilia occur around various theoretical axes of motion
- Somatic dysfunction affecting any of these axes hinders gait, requires compensatory changes, and increased energy expenditure

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DiGiovanna et al., An Osteopathic Approach to Diagnosis and Treatment, 4e, Chp. 58: Evaluation of the Pelvis

Function: Obstetrics

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The relationship of the pelvis and sacrum change throughout pregnancy.

During pregnancy:

- Center of gravity shifts anteriorly
- Sacroiliac joint and pubic symphysis become more lax (hormone: relaxin)
- ↑ Lumbar and cervical lordoses
- ↑ Thoracic kyphosis
- ↑ Anterior pelvic tilt



Somatic Dysfunction

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- What is somatic dysfunction? And what does that mean for pelvic biomechanics?
 - Impaired or altered function of related components of the body framework system: skeletal, arthrodial, and myofascial structures, and their related vascular, lymphatic, and neural elements. It is characterized by positional asymmetry, restricted range of motion, tissue texture abnormalities, and/or tenderness.

Somatic Dysfunction

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- Evaluate the sacrum and pelvis in the patient presenting with many common complaints *including but not limited to*:
 - abdominal pain, pelvic pain, dysmenorrhea, lower back pain, urinary tract complaints, lower gastrointestinal issues, neuralgia of the lower extremities, and headaches
- **Goal of treatment (OMT) is to restore *structural relationship and function* between the arthroal, neural, vascular, lymphatic, and connective tissue elements**



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https://en.wikipedia.org/wiki/Grand_Central_Terminal#/media/File:GCT_exterior.jpg



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https://en.wikipedia.org/wiki/Grand_Central_Terminal#/media/File:GCT_Aerial-Untapped_New_York-Michelle_Young.jpg

Osteopathic Structural Exam of the Pelvis

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Standing Flexion Test

+

Supine and Prone Static Examination



Somatic Dysfunction Diagnosis

Screening test # 1

Standing Flexion Test

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- Specific for ilio-sacral dysfunction
- “Pelvic somatic dysfunction”

From: **Pelvic Girdle Dysfunction**

Greenman's Principles of Manual Medicine, 5e, 2017

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Legend:
Pelvic Girdle

Standing flexion test

- Ask your patient to **stand** with his/her feet 6 - 8 inches apart
- Kneel or sit to place yourself eye-level to your patient's PSIS, and place your thumbs on the inferior aspect of the PSIS bilaterally
- Ask the patient to bend forward slowly at the waist (you may repeat to confirm)
- The test is **positive** on the side that your thumb moves "**first and farthest**"
- Name the somatic dysfunction for **the positive side**

Standing flexion test- It is your entrance to diagnosing the pelvis

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Screening test # 2

ASIS Compression Test a.k.a. Pelvic Rock Test

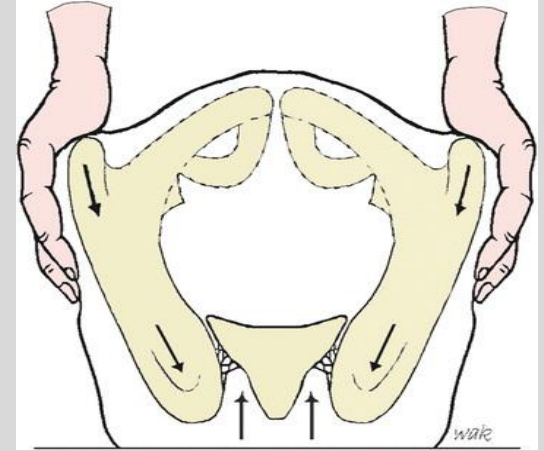
- For ilio-sacral *or* sacro-iliac dysfunction
- NOT specific for ilio-sacral dysfunction (pelvic somatic dysfunction)
 - In other words, it doesn't have to be on same side as the standing flexion test (since it can indicate a sacral SD also).



DiGiovanna et al., An Osteopathic Approach to Diagnosis and Treatment, 4e. Chp. 59: Evaluation of the Pelvis

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Seffinger MA (Exec Ed), Foundations of Osteopathic Medicine, 4e, Chp. 41B: Figure 41B.1

ASIS Compression Test a.k.a. Pelvic Rock Test

- Ask your patient to lie supine
- Place your thenar and hypothenar eminences along the patients ASISs bilaterally
- Introduce gentle but firm posterior and medial pressure toward the ipsilateral sacro-iliac joint on *one side at a time*
 - Slow
 - Allow the ilia to recoil anteriorly against gentle pressure
- Assess for *quality* of motion
 - One side usually offers more resistance
 - *Named for **side that offers more resistance***

The following test (seated flexion test) is considered specific for the sacrum (not pelvis) because it tests sacro-iliac dysfunction*

**Presented here so you understand the difference when you later learn about the sacrum in a future lecture*

Seated Flexion Test

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- Specific for sacroiliac dysfunction
- “Sacral somatic dysfunction”

Seated flexion test

- Ask your patient to **sit** with his/her feet flat on the floor and arms resting on or between the thighs
- Kneel or sit to place yourself eye-level to your patient's PSIS, and place your thumbs on the inferior aspect of the PSIS bilaterally
- Ask the patient to bend forward slowly at the waist (you may repeat to confirm)
- The test is **positive** on the side that your thumb moves **"first and farthest"**
- Name the somatic dysfunction for **the positive side**

Standing flexion test

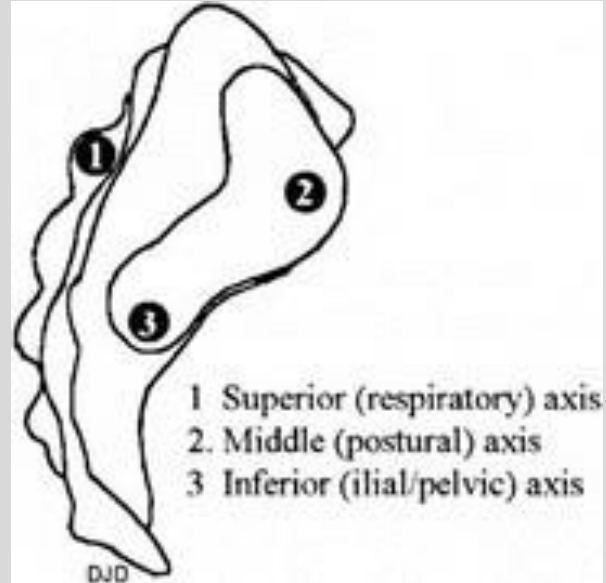
- Patient standing
- Thumbs under the PSIS
- Look for side of PSIS that moves first and farthest
- Assesses iliosacral (ie. pelvic) SD

Seated flexion test

- Patient seated
- Thumbs under the PSIS
- Look for side of PSIS that moves first and farthest
- Assesses sacroiliac (ie. sacral) SD

Iliosacral vs Sacroiliac- What's the difference?

- Iliosacral (ie. pelvic) motion refers to motion of ilium relative to the sacrum along the SI joint
 - theoretical axis of this motion is the inferior transverse axis (“3” in diagram)
- Sacroiliac (ie. sacral) motion refers to the motion of sacrum relative to ilium
 - theoretical axis of this motion is the middle transverse axis (“2” in diagram)



Seffinger, M.A. (Exec Ed), Foundations of Osteopathic Medicine, 4ed, Glossary of Osteopathic Terminology, link: <https://login.nyit.idm.oclc.org/login?url=https://osteopathicmedicine.lwwhealthlibrary.com/content.aspx?sectionid=209560259&bookid=2582#209561263>

Tips for success for the standing and seated flexion tests

1. Clearly identify the PSIS. Contacting the skin with thumbs would be ideal when you first learn this test (ask partner for permission)
2. Keep your eyes level with the PSIS; keep dominant eye at midline of body
3. Keep thumbs horizontal to the ground and inferior to each PSIS, and contact the same part of the PSIS with the same part of each thumb. **This is key step #1.**
4. When patient bends forward, keep your thumbs under the PSIS instead of letting them slide over the PSIS. **This is key step #2.**
5. Repeat your tests to ensure reliability whenever possible. **This is key step #3.**
6. Perform the ASIS compression test if the test if the above are equivocal or negative.

Pelvic Diagnosis in Clinical Practice

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- Relate it to the chief complaint
- Apply your knowledge of anatomy
- More than one dysfunction may be present
- Bilateral dysfunction may be present
- Treat and reassess somatic findings and clinical response

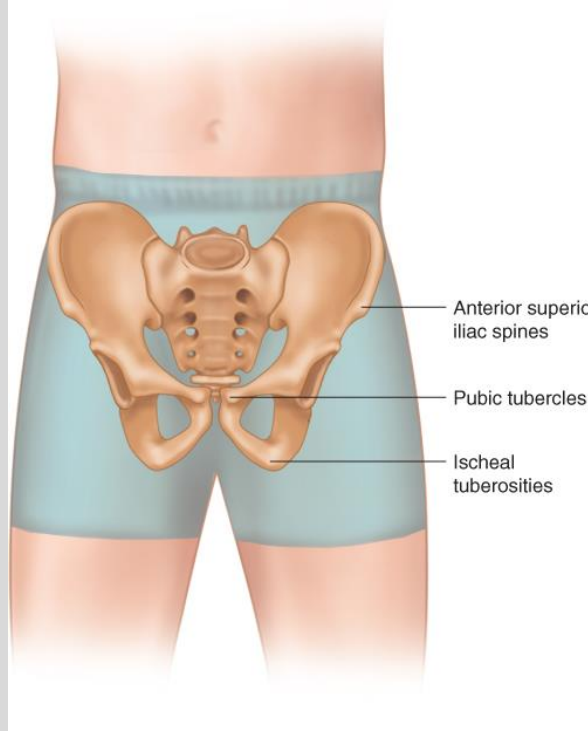
The following slides are to introduce the various landmarks that are used to diagnose pelvic somatic dysfunction. In the lab prep lecture and OMM lab for the pelvis, you will learn how to combine the tests covered in this lecture with asymmetries in key landmark findings in order to diagnose the various pelvic (ie. innominate and pubic) somatic dysfunctions. **The screening tests in this lecture as just part of the process to screen for pelvic SD. They are not sufficient for you to definitively diagnose pelvic SD. You need to examine the landmarks too!**

Static Examination of Key Pelvic Landmarks

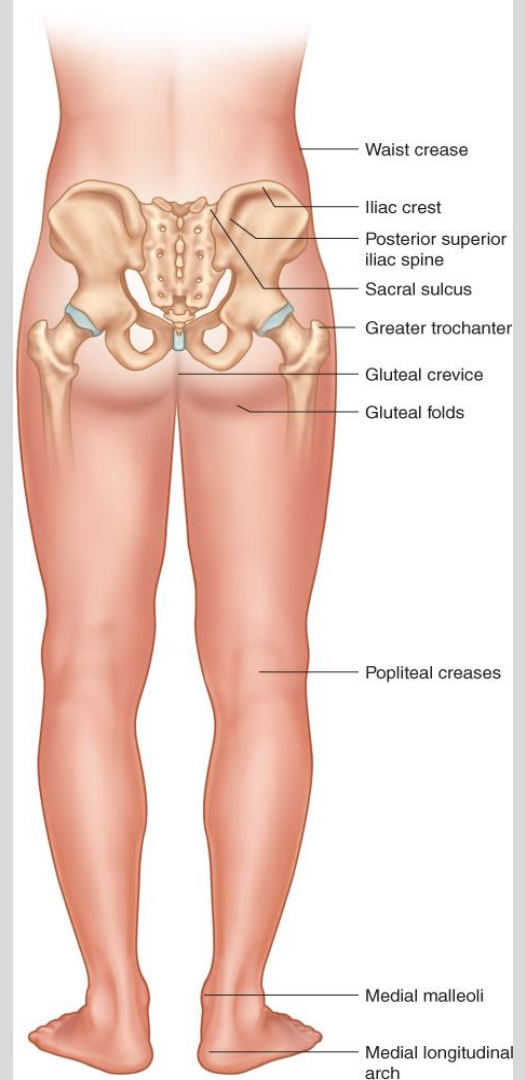
Compare Right & Left:

- ASIS heights
- PSIS heights
- Level of pubic tubercles
- Medial malleoli (not a pelvic landmark, but will learn later why it's useful)

****Make sure that person seats the pelvis first before comparing sides**



DiGiovanna et al., An Osteopathic Approach to Diagnosis and Treatment, 4e. Chp. 59: Evaluation of the Pelvis



Static Exam

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Pubic tubercles/rami



Medial Malleoli

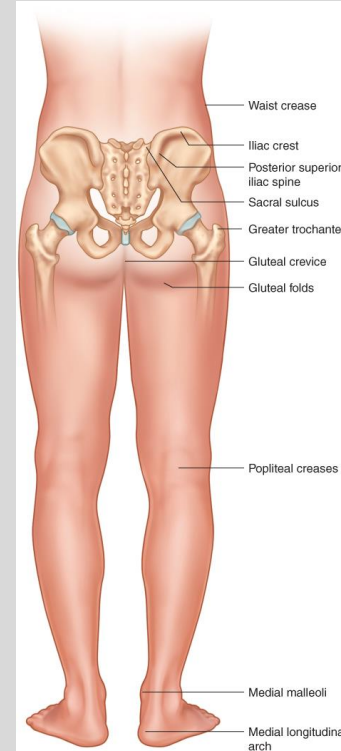


DiGiovanna et al., An Osteopathic Approach to Diagnosis and Treatment,
4e. Chp. 59: Evaluation of the Pelvis

Posterior Static Exam Landmarks

Compare Right & Left:

- Iliac Crests
- Sacral sulci
- PSIS
- ASIS (Gluteal Folds)
- Greater Trochanters
- Ischial Tuberosities
- Medial malleoli



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Please complete the following self practice questions on your own. The answers to the self-test questions will be posted at the end of the week. Please complete the questions before referring to the answer key.

Practice Question 1

- (+) Standing flexion test on R
- Which side has the pelvic somatic dysfunction?

Practice Question 2

- (+) Standing flexion test on R
 - (+) ASIS compression test on the R
 - Which side has the pelvic somatic dysfunction?
-

Practice Question 3

- (+) Standing flexion test on R
 - (+) ASIS compression test on the L
 - Which side has the pelvic somatic dysfunction?
-

Practice Question 4

- (+) Seated flexion test on R
- (+) ASIS compression test on the L
- Which side has the pelvic somatic dysfunction?

Practice Question 5

- (+) Seated flexion test on R
- (+) Standing flexion test on the R
- Which side has the pelvic somatic dysfunction?

Practice Question 6

- (+) Seated flexion test on L
- (+) Standing flexion test on the R
- Which side has the pelvic somatic dysfunction?

Practice Question 7

- (+) Seated flexion test on L
- (+) ASIS compression test on the L
- (+) Standing flexion test on the R
- Which side has the pelvic somatic dysfunction?

Practice Question 8

- (+) Standing flexion test on the L
- (-) ASIS compression test
- Which side has the pelvic somatic dysfunction?

Practice Question 9

- (+) Seated flexion test on the L
- (-) ASIS compression test
- Which side has the pelvic somatic dysfunction?

Practice Question 10

- (+) Seated flexion test on the L
 - (+) ASIS compression test on the L
 - Which side has the pelvic somatic dysfunction?
-

Practice Question 11

- (-) Seated flexion test
- (-) Standing flexion test
- (+) ASIS compression test on the L
- Which side has the pelvic somatic dysfunction?



Practice Question 12

- (-) Seated flexion test
- (-) Standing flexion test
- (+) ASIS compression test on the L and R
- Which side has the pelvic somatic dysfunction?

Practice Question 13

- (+) Seated flexion test on the L
- (-) Standing flexion test
- (+) ASIS compression test on the L and R
- Which side has the pelvic somatic dysfunction?

References

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1. DiGiovanna EL, Schiowitz S, and Dowling D 4th Ed., Wolters Kluwer 2021:

-Chapter 57: Pelvic and Sacral Anatomical Considerations, Link:

<https://login.nyit.idm.oclc.org/login?url=https://osteopathicmedicine.lwwhealthlibrary.com/content.aspx?sectionid=248813935&bookid=2969>

-Chapter 58: Gait and Postural Considerations, Link:

<https://login.nyit.idm.oclc.org/login?url=https://osteopathicmedicine.lwwhealthlibrary.com/content.aspx?sectionid=248813979&bookid=2969#248814003>

-Chapter 59: Evaluation of the Pelvis, Link:

<https://login.nyit.idm.oclc.org/login?url=https://osteopathicmedicine.lwwhealthlibrary.com/content.aspx?sectionid=248814094&bookid=2969>

2. Moore KL, Dalley AF, Agur AMR, Clinically Oriented Anatomy, 8th Ed., Wolters Kluwer 2018:

-Chapters 6 (Pelvis and Perineum), Link:

<https://login.nyit.idm.oclc.org/login?url=https://premiumbasicsscience.lwwhealthlibrary.com/content.aspx?sectionid=168704182&bookid=2212>

-Chapter 8 (Lower Limb), Link:

<https://login.nyit.idm.oclc.org/login?url=https://premiumbasicsscience.lwwhealthlibrary.com/content.aspx?sectionid=168705038&bookid=2212>

3. Seffinger, Michael A., D.O., executive editor, Foundations of Osteopathic Medicine, Philosophy, Science, Clinical Applications, and Research, Fourth Edition, Wolters Kluwer, 2018:

-Chapter 8: Anatomy and Physiology of the Lymphatic System, Link:

<https://login.nyit.idm.oclc.org/login?url=https://osteopathicmedicine.lwwhealthlibrary.com/content.aspx?sectionid=209540059&bookid=2582#209540161>

-Chapter 41B: Diagnosis and Treatment of Somatic Dysfunction in the Patient with Low Back Pain, Link:

<https://login.nyit.idm.oclc.org/login?url=https://osteopathicmedicine.lwwhealthlibrary.com/content.aspx?bookId=2582§ionId=209552184&resultClick=1#209552244>

4. Pixabay.com

Lecture and Lab Feedback Form:

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